Project Sentinel · Official Submission Manifest

Cryptographic Package Verification & Component Functional Index

Kempower Challenge · LahtiES Energy Tech Hackathon 2026 (LUT University Lahti / CrazyTown)

- **Package Archive**: `Team3_Project_Sentinel_Submission.zip` (also accessible as `submission.zip`)
- **Total Deliverable Files**: 177 (including this manifest & its companion PDF)
- **Total Uncompressed Volume**: 28.47 MB (29,850,931 bytes)
- **Zero-CJK Rule Compliance**: 100% Pure Professional English Verified (`[\u4e00-\u9fff]` match count = 0)
- **Deterministic Verification**: `python 2_The_Demonstration_Part2/verify_all.py` (29 checks passing in < 0.5s)
- **Automated Test Coverage**: 462 Pytest + 106 E2E = 568 total automated tests (100% pass rate)

Deliverable Summary by Pillar

- `root/` (Top-level documentation & manifest): 2 files (607.67 KB)
- `1_pitch_and_story_2029/` (Saturday Deliverable 1: 10-Minute Pitch & 2029 Customer Story): 11 files (17.79 MB)
- `2_demonstration_part2/` (Saturday Deliverable 2: The Demonstration (Scenario Day 13.11.2026 Engine)): 106 files (1.91 MB)
- `3_commercial_and_one_page_sla/` (Saturday Deliverable 3: Commercial Model & One-Page Service Agreement): 8 files (1.01 MB)
- `4_energy_and_control_core/` (Challenge Part 1: Energy & Control Core Architecture & Audits): 48 files (7.16 MB)

Complete Cryptographic Inventory

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum
1	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/CUSTOMER_STORY_2029.md	6.13 KB	707c66a792dd8a72ac607c496f7ba2
2	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/CUSTOMER_STORY_2029.pdf	113.56 KB	15e3fd64ac207c80d16342e528eb02
3	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/PITCH_DECK_OUTLINE_v2.md	44.95 KB	204e2a9c5640f609e12c9908903296
4	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/PITCH_DECK_OUTLINE_v2.pdf	667.68 KB	7df7acfad2357584064e109aae07d3
5	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/PPT_SPEECH_PLAN.md	33.13 KB	a6ce68c132c0a78c6f99a236924df6
6	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/PPT_SPEECH_PLAN.pdf	461.51 KB	e4982f6e49c308d77986af97be1965
7	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/Project_Sentinel_12pages_v3.pptx	16.06 MB	9582147dc0ea0de3d668521334c68c
8	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/Project_Sentinel_Pitch_Script_12pages.md	14.79 KB	7a684095e1e6e66e6576a33090932b
9	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/Project_Sentinel_Pitch_Script_12pages.pdf	169.58 KB	e0365c65335a44fce5f18f599037ca
10	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/SPEECH_10MIN_ENGLISH_OFFICIAL.md	24.88 KB	164dacc7c1fa60cb006b04a51b1c1d
11	1_pitch_and_story_2029	1_Pitch_and_2029_Customer_Story/SPEECH_10MIN_ENGLISH_OFFICIAL.pdf	232.94 KB	b636e05c3041f849995e10b251edeb
12	2_demonstration_part2	2_The_Demonstration_Part2/DEMONSTRATION_WALKTHROUGH.md	6.61 KB	45e53483250d8a88c5b8489313fc29
13	2_demonstration_part2	2_The_Demonstration_Part2/DEMONSTRATION_WALKTHROUGH.pdf	152.15 KB	b40379012b0d75b8aee63d6dff931f
14	2_demonstration_part2	2_The_Demonstration_Part2/SPEECH_OVERHEAT_SCENE.md	9.17 KB	d42e7d94fc6c27176052d87b432aa8
15	2_demonstration_part2	2_The_Demonstration_Part2/SPEECH_OVERHEAT_SCENE.pdf	151.03 KB	91709a09953ec04abd22d02c0deb38
16	2_demonstration_part2	2_The_Demonstration_Part2/SPEECH_SIX_SCENES.md	12.60 KB	6beabed6486dd43476d4c3bcad97c0
17	2_demonstration_part2	2_The_Demonstration_Part2/SPEECH_SIX_SCENES.pdf	144.99 KB	5a7ff6a119b17626bc5cbb3ca47e38
18	2_demonstration_part2	2_The_Demonstration_Part2/app.py	32.36 KB	912a1c2e7c073e003ba49a57c8cf91
19	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/.streamlit/config.toml	235 B	34824b6fab69a5c93cd790e1c9180c
20	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/README.md	2.50 KB	a8adfad81f84c4e02bd6a761186189
21	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/README.pdf	96.94 KB	a54ef0f35311a4e46e2de4e6c83f27
22	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/SPEECH_OVERHEAT_SCENE.md	9.17 KB	d42e7d94fc6c27176052d87b432aa8
23	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/SPEECH_OVERHEAT_SCENE.pdf	151.03 KB	66434d4fc6e6e666b5b109ce430a29
24	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/SPEECH_SIX_SCENES.md	12.60 KB	6beabed6486dd43476d4c3bcad97c0
25	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/SPEECH_SIX_SCENES.pdf	144.99 KB	b17819b71b89799cc2be74c8d5c01f
26	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/app.py	9.78 KB	7d244b2243656319f1f1cccff8b984
27	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/pages/1_Night_Replay.py	16.84 KB	53a1fa5ef00026ec35a7af4aeebb64
28	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/pages/2_Outage_Replay.py	15.19 KB	2af9ddec95691b1e0fbe1eedb1c8c0

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum
29	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/pages/3_Price_Spike_Replay.py	3.16 KB	6ccbed7c0a862a8510c4b1fd8a3311
30	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/pages/4_Curtailment_Replay.py	3.14 KB	bbb06054f14053392b36b7aebccc51
31	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/pages/5_Cable_Fault_Replay.py	3.22 KB	af44344ba7e4cf3c06acba2b351c6c
32	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/pages/6_Commissioning_Replay.py	3.07 KB	1c757b29c09d7cb2ae36dbc503862f
33	2_demonstration_part2	2_The_Demonstration_Part2/demo_scenarios/requirements.txt	491 B	89fcc60e66fbc706deeaaa9777cc63
34	2_demonstration_part2	2_The_Demonstration_Part2/e2e_tests/__init__.py	90 B	de00142fb1e50bcd4418d8b2c7263a
35	2_demonstration_part2	2_The_Demonstration_Part2/e2e_tests/common.py	4.82 KB	828699c80d91ea344d6ba0ad9b8c53
36	2_demonstration_part2	2_The_Demonstration_Part2/e2e_tests/test_runner.py	5.79 KB	1dcf1cb66c6df0fb0eaa57333b1dae
37	2_demonstration_part2	2_The_Demonstration_Part2/e2e_tests/tier1_feature_coverage.py	20.12 KB	8223a3139b255427e58ffd605b7125
38	2_demonstration_part2	2_The_Demonstration_Part2/e2e_tests/tier2_boundary_corner.py	13.35 KB	19f07c20f80a3eafeaaabace23545f
39	2_demonstration_part2	2_The_Demonstration_Part2/e2e_tests/tier3_cross_feature.py	6.44 KB	4542d7bc49d5805f984319d3509435
40	2_demonstration_part2	2_The_Demonstration_Part2/e2e_tests/tier4_real_world_scenario.py	8.69 KB	37576eb3980aa71881606eab7b3668
41	2_demonstration_part2	2_The_Demonstration_Part2/pytest.ini	42 B	b160d82f7811555051f8966d8e0275
42	2_demonstration_part2	2_The_Demonstration_Part2/requirements.txt	488 B	31e6ed5500590b635780438f395a0d
43	2_demonstration_part2	2_The_Demonstration_Part2/run_s1.py	10.13 KB	e0fef5a26ad049e0890b3d6331705e
44	2_demonstration_part2	2_The_Demonstration_Part2/src/__init__.py	64 B	1abc659bc8548aaa23a37b1b161f89
45	2_demonstration_part2	2_The_Demonstration_Part2/src/commercial/__init__.py	253 B	92bf037063f5bc20ddfdb80cfd5865
46	2_demonstration_part2	2_The_Demonstration_Part2/src/commercial/roi_analyzer.py	10.30 KB	a3d8f7a36b1d8de4156579e8b34406
47	2_demonstration_part2	2_The_Demonstration_Part2/src/commercial/sensitivity_analyzer.py	10.88 KB	ed1f8f98fbaffe9d90c5f8d7e24ea0
48	2_demonstration_part2	2_The_Demonstration_Part2/src/commercial/settlement_clearing.py	18.15 KB	515f22d2f0ebf93ea465a86190444a
49	2_demonstration_part2	2_The_Demonstration_Part2/src/commercial/sla_calculator.py	4.41 KB	176b755b5e9395ba85653ba7bf7546
50	2_demonstration_part2	2_The_Demonstration_Part2/src/core/__init__.py	1.99 KB	3ec1b6a99e8e63783cfaae64afcc4c
51	2_demonstration_part2	2_The_Demonstration_Part2/src/core/anti_fraud.py	9.54 KB	17f392630087311f25aae723ab44f4
52	2_demonstration_part2	2_The_Demonstration_Part2/src/core/dpr_matrix.py	29.74 KB	139fffd4616829f39f49a4128ead4c
53	2_demonstration_part2	2_The_Demonstration_Part2/src/core/mpc_optimizer.py	21.21 KB	b0a4daee6238ab1d0aea01a9abc700
54	2_demonstration_part2	2_The_Demonstration_Part2/src/core/protocols.py	40.56 KB	476f7ef80ba7ea7546a50fbe7a481c
55	2_demonstration_part2	2_The_Demonstration_Part2/src/core/self_healing.py	40.85 KB	1cfd3757821f355631abbd793ff61b
56	2_demonstration_part2	2_The_Demonstration_Part2/src/core/virtual_replug.py	25.39 KB	a7fb6e1f558e6325d89d1b05c71b68
57	2_demonstration_part2	2_The_Demonstration_Part2/src/edf_packer.py	10.87 KB	2c77ed8343639b39f75386842d1191
58	2_demonstration_part2	2_The_Demonstration_Part2/src/sim/__init__.py	265 B	879039c8c27764d2aa6b76ed9c796c
59	2_demonstration_part2	2_The_Demonstration_Part2/src/sim/data_loader.py	4.82 KB	89cacb4f2df1397b1cb9b3088ad55d

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum
60	2_demonstration_part2	2_The_Demonstration_Part2/src/sim/discrete_sim.py	24.13 KB	ead22a7ccb56c684923ebca0b7c69f
61	2_demonstration_part2	2_The_Demonstration_Part2/src/sim/fault_replays.py	24.68 KB	e38becc1669acbbc35f499822b9ef2
62	2_demonstration_part2	2_The_Demonstration_Part2/src/sim/night_replay.py	15.50 KB	473d1b8de78046cec9b1b8911e7284
63	2_demonstration_part2	2_The_Demonstration_Part2/src/sim/outage_replay.py	9.24 KB	57e60806400e5e2d1c0033994a29b6
64	2_demonstration_part2	2_The_Demonstration_Part2/src/sim/scenario_engine.py	16.22 KB	1c12c39430a5c7f74444f7425bae10
65	2_demonstration_part2	2_The_Demonstration_Part2/src/ui/__init__.py	84 B	81d9ba7d47beb44af1384215a904f7
66	2_demonstration_part2	2_The_Demonstration_Part2/src/ui/replay_scene.py	22.61 KB	bd6a0fde23441b7f4af13c623e98fc
67	2_demonstration_part2	2_The_Demonstration_Part2/src/utils/__init__.py	211 B	f96583f042dbc2fc7cab4bf9558fb4
68	2_demonstration_part2	2_The_Demonstration_Part2/src/utils/paths.py	1.57 KB	7d219ac723250015622e02d9134479
69	2_demonstration_part2	2_The_Demonstration_Part2/tests/analyze_scenario_day.py	3.19 KB	9b8d5c9b34b6655cd2adb199ec55b3
70	2_demonstration_part2	2_The_Demonstration_Part2/tests/check_bess.py	554 B	6c6f5f1532f1b68f4f75b03f32ef62
71	2_demonstration_part2	2_The_Demonstration_Part2/tests/check_d1_9.py	374 B	415358dd9fef8a0d4e93348461592a
72	2_demonstration_part2	2_The_Demonstration_Part2/tests/check_d1_later.py	878 B	bda9828a9aee79bb65f99ca0598f97
73	2_demonstration_part2	2_The_Demonstration_Part2/tests/check_s1_power.py	704 B	cb16ddba8d133dedf79ae9942d3770
74	2_demonstration_part2	2_The_Demonstration_Part2/tests/check_s3_outage.py	875 B	91f87cf90aa8b92cced44c86f0f348
75	2_demonstration_part2	2_The_Demonstration_Part2/tests/check_t01_t04.py	404 B	e3ed4d5f76d62a9bb5dc632d71d9b6
76	2_demonstration_part2	2_The_Demonstration_Part2/tests/conftest.py	157 B	f1efac1390e32ed1e95fc156828ccb
77	2_demonstration_part2	2_The_Demonstration_Part2/tests/edf_simulation.py	3.39 KB	6b1952c1a6548386e8446a7bb7e35b
78	2_demonstration_part2	2_The_Demonstration_Part2/tests/inspect_cols.py	454 B	8be91976946d4e3a19421fe3f516d2
79	2_demonstration_part2	2_The_Demonstration_Part2/tests/inspect_dataset.py	955 B	4d3442437c17b4eb003a322ca67f54
80	2_demonstration_part2	2_The_Demonstration_Part2/tests/inspect_s1_telemetry.py	1.10 KB	f06cdaabdd4f8db9305f2b164e6950
81	2_demonstration_part2	2_The_Demonstration_Part2/tests/query_scenario_data.py	1.53 KB	a5c848437b5e132665655201603f6e
82	2_demonstration_part2	2_The_Demonstration_Part2/tests/simulate_s2_curtailment.py	2.75 KB	b88a98fb37592964b92d3802faa6ee
83	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_adversarial_discrete_sim_thermal.py	24.76 KB	123976b1868da5328a43f6cbfbe09d
84	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_adversarial_m1.py	19.24 KB	3d22b03e00023d1c208f3dd876876b
85	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_anti_fraud_replug.py	5.19 KB	93b485cc6dfc88184f95686be1c720
86	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_commercial_settlement.py	15.60 KB	4c73d7bdd9fec16867f1151098a993
87	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_dashboard_runtime_stress.py	14.51 KB	ce3f40c9a5c25b6d60c1baff3dd868
88	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_demo_day_simulation.py	19.28 KB	bbb4afbd86d92b222b1ef4cd8ed58a
89	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_discrete_sim_engine.py	8.00 KB	c6012da71b0f8dc933557e9447236a
90	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_dpr_stress_adversarial.py	19.89 KB	666effd97188e1f1534e94ff1fd7e1
91	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_e2e_suite.py	45.75 KB	10b7ad1b2b5c69f94687a08ac158bf

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum
92	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_gate2_empirical_verification.py	6.97 KB	c4a1e8f11cc9101874febb8a21b534
93	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_github_issue2_29items.py	14.33 KB	243ef14cb284c97473a0a12f0ce3a5
94	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_m1_adversarial_probe.py	21.96 KB	879b0152b52209f98b372a0ea5860b
95	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_m1_remediation.py	13.75 KB	c408e705ba07c7f3f8aa9998e9bb73
96	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_m2_challenger_empirical.py	21.39 KB	3daf44cf881c14f7532e2966fdd5e0
97	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_m2_hardware_edge_stress.py	17.97 KB	f7118be5fb53d1be6ab324d9b968d7
98	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_m2_hardware_self_healing.py	16.38 KB	ab2808b81adaad14d89ae56bd5c73e
99	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_m2_protocols_anti_fraud_adversarial.py	19.76 KB	0370eeef177b3a24cd67b85513649d
100	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_m3_adversarial_stress.py	16.52 KB	218df08d523ca666d4437d6e0351bd
101	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_mpc_math_fixes.py	5.64 KB	a721eaf0bb90da158a2e37d91ae2c3
102	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_protocols_ocpp_iso.py	14.93 KB	dbdddd0672d752b41e1f95372a6828
103	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_remediated_sla.py	5.71 KB	b304876d5bd0d4742c288319e51963
104	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_sla_edge_cases.py	3.77 KB	92732951bf0c93e53350d4c8b28e4b
105	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_tier5_commercial_adversarial.py	42.05 KB	c3f3871bf19d77315e2a5ab9d9dc33
106	2_demonstration_part2	2_The_Demonstration_Part2/tests/test_tier5_core_adversarial.py	43.93 KB	a0838458f9ae65f9184428aea32a84
107	2_demonstration_part2	2_The_Demonstration_Part2/tests/verify_csv_inventory.py	2.94 KB	4c97df1cf3dd36ae7cb8aac3c9ea9b
108	2_demonstration_part2	2_The_Demonstration_Part2/tests/verify_empirical_claims.py	7.84 KB	c22ca88a0dd01ca42b8601e615e75a
109	2_demonstration_part2	2_The_Demonstration_Part2/tools/check_demo_artifacts.py	3.12 KB	7f35cdc91661faece79eaf71b2536d
110	2_demonstration_part2	2_The_Demonstration_Part2/tools/cjk_audit.py	1.41 KB	3ba5b8cd1979b16a15df80afa705f1
111	2_demonstration_part2	2_The_Demonstration_Part2/tools/export_deck_ledger.py	39.59 KB	a20260946a57c8857add3b5e3d5b70
112	2_demonstration_part2	2_The_Demonstration_Part2/tools/md_to_pdf.py	5.74 KB	02a2f804faa6655534301dcf6639ba
113	2_demonstration_part2	2_The_Demonstration_Part2/tools/package_and_manifest.py	16.31 KB	2d2381e9912233f1589d5ba69c2737
114	2_demonstration_part2	2_The_Demonstration_Part2/tools/s1_multicabinet_sim.py	12.12 KB	ea9c9c403c849dd14cecfb3ea9435c
115	2_demonstration_part2	2_The_Demonstration_Part2/tools/stage_submission.py	7.97 KB	83cbb34d555b691706bfa24c56ffa0
116	2_demonstration_part2	2_The_Demonstration_Part2/tools/verify_submission_cjk.py	2.33 KB	b142ee1ac6b2c129459ca444d7e793
117	2_demonstration_part2	2_The_Demonstration_Part2/verify_all.py	6.76 KB	f8d05f4dd439d48773ad4aafdd4346
118	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/COMMERCIAL_BUSINESS_MODEL.md	4.97 KB	61d6db3b85b684af95aee906ffaa23
119	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/COMMERCIAL_BUSINESS_MODEL.pdf	116.12 KB	3dd0652f8444ea1efa9a6bfbd541c6

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum
120	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/One_Page_SLA_Specification.md	22.20 KB	65fe05c7e3269eea74359e9f966929
121	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/One_Page_SLA_Specification.pdf	634.92 KB	2de6c344ae84c071c95a582be52291
122	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/Team3-One_Page-Autonomous_Charging_Services_Agreement.docx	92.76 KB	0158958048cac203dd591ce589bf04
123	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/one_page_agreement.html	11.08 KB	a92cb6b5c7427ea02d216cd5d97ea4
124	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/one_page_agreement.md	6.58 KB	254e3961995d99d9d151b2db2f824a
125	3_commercial_and_one_page_sla	3_Commercial_Model_and_One_Page_SLA_Part3/one_page_agreement.pdf	148.54 KB	1704d5fd17544d91ee0e01f2fb99a0
126	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/HACKATHON.md	3.72 KB	be16e96cf571ffabb8f728d489dc26
127	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/HACKATHON.pdf	152.82 KB	3c59d345cc79ab2bd9de50a6c0e11a
128	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/PROJECT.md	17.15 KB	1804e340c355504f911c24279dc490
129	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/PROJECT.pdf	224.51 KB	9ef329031b00ee69da26cdb93debf6
130	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/PROJECT_AUDIT_REPORT.md	7.17 KB	9ee879203249404d8ac5b1c49ea2be
131	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/PROJECT_AUDIT_REPORT.pdf	142.42 KB	0c9ac24766ed8c5638c4d5c9840325
132	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/README_EN.md	34.43 KB	b376c662d66faa51dcccdb4e22471a
133	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/README_EN.pdf	510.10 KB	34d30c855450304c0f5e4ef1a12a48
134	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/ZERO_HUMAN_DESIGN.md	18.70 KB	4a1df3da519f8b69c763474128e7fc
135	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/ZERO_HUMAN_DESIGN.pdf	312.53 KB	94fb88b0b533ac305be258980b08ae
136	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/assumptions_and_additions.md	11.67 KB	b51226c0eabfe14f722e204828169e
137	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/assumptions_and_additions.pdf	277.50 KB	4738beaa764af89968d3bcb2d789f8
138	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/attribution_matrix.md	13.94 KB	c589f83d68fa7cfcada147a9c1c7e4
139	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/attribution_matrix.pdf	466.86 KB	6ba7c7c239ade6df9b405c8cdccc87
140	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/00_pocket_cheat_sheet.md	8.88 KB	8c1e41f37a71acf3d5591d7c2eaed0
141	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/00_pocket_cheat_sheet.pdf	157.29 KB	61294af0ec2b27cd213c25a255eff0
142	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/01_stage_schedule_and_setup.md	47.69 KB	94bd789edce7e64106fb7bb345c0c8
143	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/01_stage_schedule_and_setup.pdf	433.11 KB	ed98c192deefd177f00509b0612309
144	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/02_team_roles_and_coordination.md	32.92 KB	35a0133c5902c568cfdc30d3084817
145	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/02_team_roles_and_coordination.pdf	367.02 KB	3a00350da876bc72eaf24fcc59899a
146	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/03_judge_personas_and_question_bank.md	31.10 KB	fbe274474571ad072923fd77e3f4a0
147	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/03_judge_personas_and_question_bank.pdf	305.30 KB	2b872d0b6621383906fb4023043ca0

#	Pillar Category	Relative File Path	Size	SHA-256 Checksum
148	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/04_structured_defense_and_crisis_manual.md	48.96 KB	bfde4d27f643c05125d413f31e3906
149	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/04_structured_defense_and_crisis_manual.pdf	571.60 KB	52e641b0f9659f072e5ab236e80b99
150	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/05_live_demo_operational_sop.md	12.56 KB	d80535b094e780b7ba1b0c5dfe0cf0
151	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/05_live_demo_operational_sop.pdf	288.99 KB	b6d8f510091f9432b1017d2c8736d2
152	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/06_code_proof_and_authenticity_guide.md	12.18 KB	1aafcb5d859dae4de823a21f10d3a0
153	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/06_code_proof_and_authenticity_guide.pdf	403.17 KB	143b6734ecdbfdc010c43e47141e19
154	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/PROJECT_SENTINEL_MASTER_PORTRAIT_AND_FILE_INDEX_EN.md	34.43 KB	b376c662d66faa51dcccdb4e22471a
155	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/PROJECT_SENTINEL_MASTER_PORTRAIT_AND_FILE_INDEX_EN.pdf	510.10 KB	29befa61bed349efdb8270ca25ccb7
156	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/README.md	16.53 KB	14e87f49a9ca2ea5871ea0a355b675
157	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/demo_day/README.pdf	180.53 KB	ef4688c5aae06145c33d0661623f42
158	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/energy_impact.md	20.66 KB	deffb9be6a910a30dc7f277e1bca72
159	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/energy_impact.pdf	536.04 KB	ef3a008da7a5015b39b3f0876caa6c
160	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/BRIEF.md	6.18 KB	571c27eb8738cb0b1a6d1ce3775c6a
161	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/BRIEF.pdf	193.66 KB	d832364f20f718b8d25b1d5ce0928d
162	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/BUILD_PLAN.md	2.31 KB	11af160f1836588db876ec8d590c70
163	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/BUILD_PLAN.pdf	115.57 KB	51012e7c0316dd3627204e1069586d
164	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/DECISIONS.md	3.41 KB	db08a4e7f78a9724bc30a86188fabc
165	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/DECISIONS.pdf	156.95 KB	ffe20f0d5fb925a5d5d1281997ac5e
166	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/DEMO_SCRIPT.md	5.61 KB	4b3979ec02a20e050d04812cc9ab0f
167	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/DEMO_SCRIPT.pdf	138.88 KB	7fe70c461b6bb77bf448c8fda24e1f
168	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/JUDGING.md	5.51 KB	c6b827e9d0665a249fc563c9e00e0c
169	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/JUDGING.pdf	173.19 KB	2144fd3d01f4649fea1a34e1635fad
170	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/RULES.md	3.51 KB	76170f3d29259f6c0232486e31fbc9
171	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/RULES.pdf	135.55 KB	a0c5513a47c49e76380a381e50f790
172	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/SUBMISSION.md	5.32 KB	13f6abe5fee18c06ecbc981db851a4
173	4_energy_and_control_core	4_Energy_and_Control_Core_Part1/hackathon/SUBMISSION.pdf	177.94 KB	dcc48bcac2d5842fb09095ea086670
174	root	README.md	24.05 KB	d1433f2282812746cfeb884e428e96
175	root	README.pdf	583.62 KB	b3ba0aa898ede7617e1b7017a3ee1d

Independent Verification Instructions

To independently verify the integrity of all files in this submission, run:

```
# 1. Unzip the official archive
unzip Team3_Project_Sentinel_Submission.zip -d submission_verify/
cd submission_verify/2_The_Demonstration_Part2/

# 2. Run the deterministic verifier (< 0.5s)
python verify_all.py

# 3. Run the complete E2E and Pytest test suites
python -m e2e_tests.test_runner
pytest tests/ -q
```